

levels. If resolution is 0.1 dB, power levels separated by less than 0.1 dB cannot be distinguished. In case, the stability of power amplifiers is 0.2 dB/day, to measure it, the power meter should have a resolution higher than 0.2 dB.

Other specifications

Rise time. Some specifications are relevant to a particular type of measuring instruments. For example, rise time is relevant to oscilloscopes. For pulse waveform measurements, rise-time specification is very important. It is defined as the time taken to reach from 10 per cent to 90 per cent of the maximum value. Rise time depends on the bandwidth of the oscilloscope:

Rise time (in nanoseconds)=350/
Bandwidth (in megahertz)

A 100MHz oscilloscope will have a rise time of 3.5 nanoseconds. For analysing fast-rising waveforms, oscilloscopes should have a fast rise time. The measured rise time, the oscilloscope rise time and the actual rise time are related by:

$$t_{\text{measured}} = \sqrt{t_{\text{osc}}^2 + t_{\text{waveform}}^2}$$

Thus for minimal measurement error, the oscilloscope rise time should

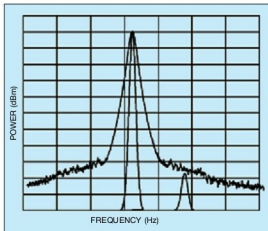


Fig. 2: Because of high FM noise, noise level close to the carrier is very high, masking the low-level signal

be small compared to that of the waveform.

FM noise. This specification is important for frequency synthesisers. Phase-locked oscillators are used for carrier generators in communication links. Synthesisers are used as the reference source for phase-locked oscillators. If multiplication factor is 'N,' FM noise after multiplication will increase by $20 \log N$. In higher FM noise, carriers may overlap with adjacent channels. Also, in narrowband communications, this will affect signal detection.

Hence FM noise in a frequency synthesiser should be properly specified. It is specified as single-sideband phase noise and given as X dBc/Hz at Y Hz away from the carrier. This means the phase noise is X dB below the carrier frequency at Y Hz away from the carrier frequency in a measurement bandwidth of 1 Hz.

In spectrum analysers also, phase noise is important as it determines purity of the measured signal. Also, unequal signals cannot be resolved as low-level signals will merge with the noise.

In Fig. 2, because of high FM noise, noise level close to the carrier is very high, masking the low-level signal. Thus noise sidebands affect the resolution of low signals close to the carrier.

General specifications. General specifications are common to all types of instruments. For example, GPIB/RS-232 interface is needed to connect the instrument with a PC for control and status monitoring. The instruments can be controlled remotely through the PC and all the front-panel readings monitored on the PC. Power source specification and environmental specifications like operating temperature and relative humidity are also common to all instruments.

Table I shows major specifications that are applicable to various measuring instruments. 'Yes' means the particular specification is applicable for the instrument and 'No' means the specification is not applicable.

In oscilloscopes, where CRT read-out is available, frequency resolution may be specified. In instruments like spectrum analysers, power meters, frequency counters and oscilloscopes, where signal is fed to the instrument, input impedance is specified. Whereas in instruments like generators/frequency synthesisers, where output is taken, output impedance is specified.

TABLE I
Major Specifications Applicable to Various Instruments

Specifications	Instruments				
	Spectrum analyser	Generator/frequency synthesiser	Power meter	Frequency counter	Oscilloscope
Frequency range	Yes	Yes	Yes	Yes	Yes
Frequency resolution	Yes	Yes	No	Yes	No
Frequency accuracy	Yes	Yes	No	Yes	No
Sensitivity	Yes	No	Yes	Yes	Yes
Maximum input	Yes	No	Yes	Yes	Yes
Level accuracy	Yes	Yes	Yes	No	Yes
Rise time	No	No	No	No	Yes
Phase noise	Yes	Yes	No	No	No
Input impedance	Yes	No	Yes	Yes	Yes
Output impedance	No	Yes	No	No	No
Frequency response	Yes	Yes	Yes	No	Yes
Third-order intermod	Yes	No	No	No	No
Power source, operating temperature, humidity, etc	Yes	Yes	Yes	Yes	Yes
Recorder/printer output facility	Yes	No	Yes	Yes	Yes